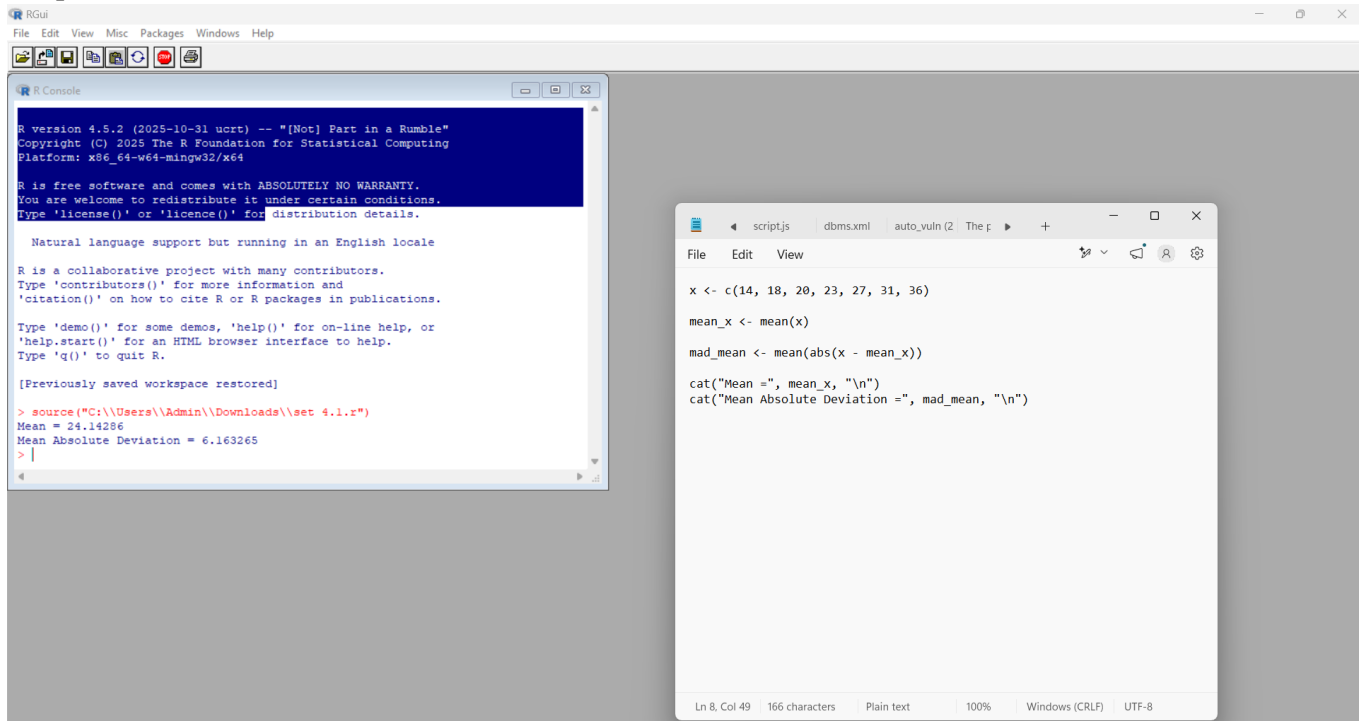


SET 4

1. R: Calculate mean absolute deviation from the mean for 14,18,20,23,27,31,36.

Output:



The screenshot shows the RGui interface. The R Console window on the left displays the R startup message and the results of running the script. The script editor on the right shows the R code used to calculate the mean and mean absolute deviation.

```
R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Rumble"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]
> source("C:\\Users\\Admin\\Downloads\\set 4.1.r")
Mean = 24.14286
Mean Absolute Deviation = 6.163265
> |
```

```
x <- c(14, 18, 20, 23, 27, 31, 36)

mean_x <- mean(x)

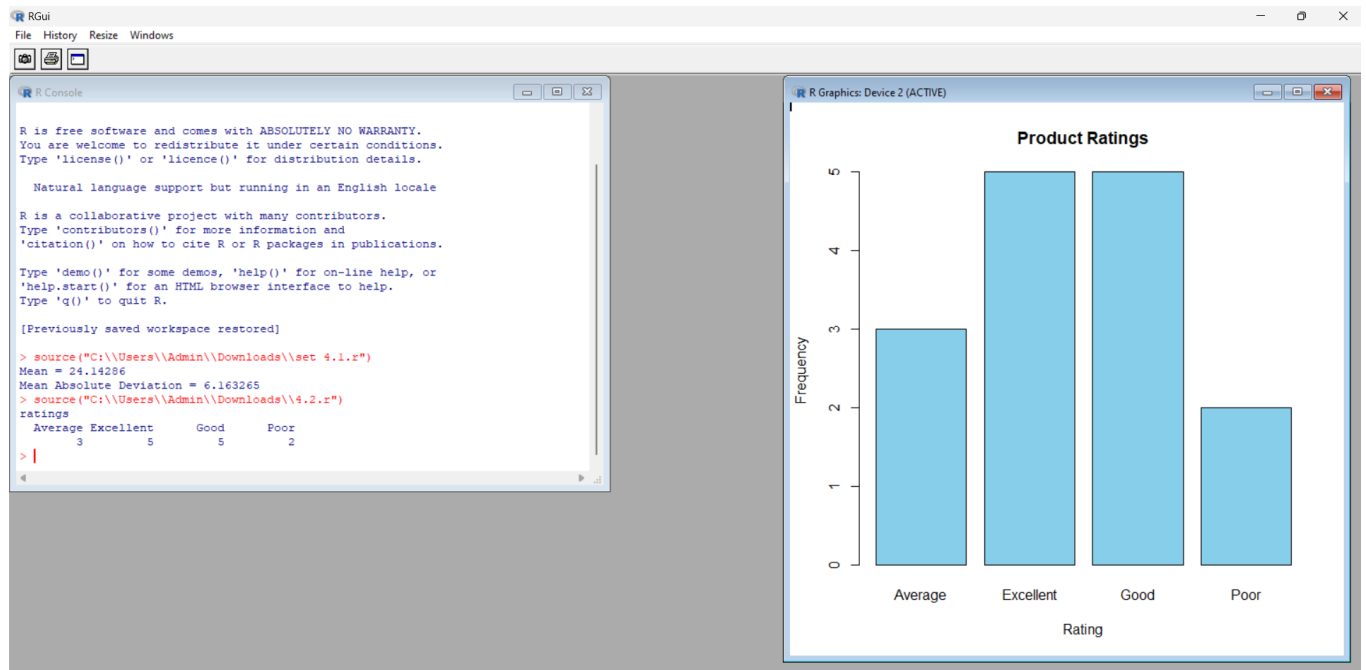
mad_mean <- mean(abs(x - mean_x))

cat("Mean =", mean_x, "\n")
cat("Mean Absolute Deviation =", mad_mean, "\n")
```

Ln 8, Col 49 166 characters Plain text 100% Windows (CRLF) UTF-8

2. R: Create 15 categorical product ratings; generate a frequency table and bar plot.

Output:

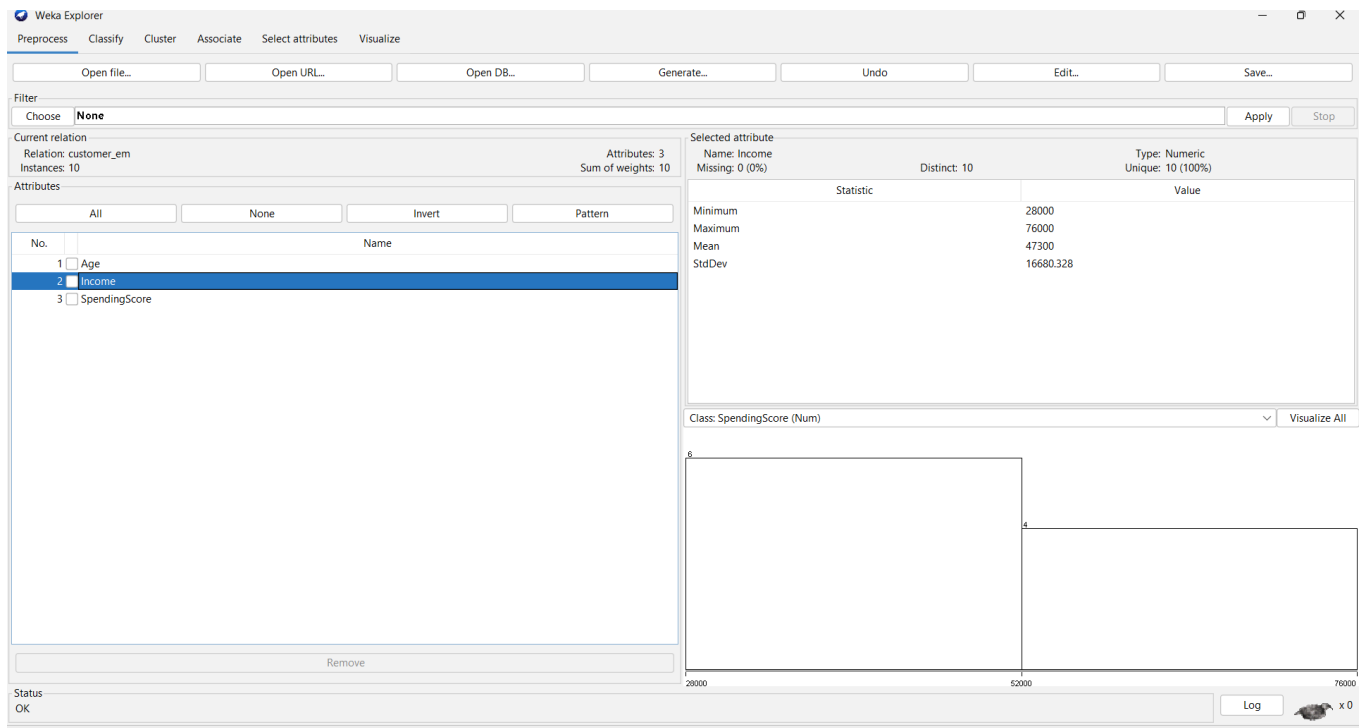


3. WEKA: Apply EM clustering to a suitable numeric dataset and compare its clusters with SimpleKMeans.

Dataset: EM Clustering

```
@relation customer_em
@attribute Age numeric
@attribute Income numeric
@attribute SpendingScore numeric
@data
22,28000,35
25,32000,40
27,35000,45
30,42000,55
34,50000,60
38,58000,70
42,68000,78
46,76000,85
24,30000,38
36,54000,65
```

Output:



4. WEKA: Apply COBWEB to the same dataset and inspect the hierarchical cluster structure.

Dataset: COBWEB

```
@relation customer_hierarchical
@attribute Region {North,South,East,West}
@attribute Visits numeric
@attribute Purchases numeric
@attribute Rating numeric
@data
North,12,3,3
North,18,5,4
South,9,2,2
South,14,4,3
East,22,7,5
East,20,6,4
West,8,1,2
West,11,3,3
```

Output:

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Open file...Open URL...Open DB...Generate...UndoEdit...Save...

FilterChooseNoneApplyStop

Current relation
Relation: customer_hierarchical
Instances: 8

Attributes: 4
Sum of weights: 8

Attributes

AllNoneInvertPattern

No.	Name
1	Region
2	Visits
3	Purchases
4	Rating

Remove

Selected attribute
Name: Region
Missing: 0 (0%)

Distinct: 4

Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	North	2	2
2	South	2	2
3	East	2	2
4	West	2	2

Class: Rating (Num)Visualize All

2

2

2

2

South [2]

Status
OK

Log x 0

